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Big Five personality traits and university students' academic performance: A meta-analysis

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ABSTRACT

Meta-analytical evidence on the linkage between university students' academic performance and personality traits, considering cultural and contextual factors, remains limited. This meta-analysis reports comprehensive assessment of the relationships between personality traits and academic performance, incorporating 84 studies published between 1995 and 2024. A total of 370 correlations with a combined sample size of 46,729 were analyzed. Academic performance was found to correlate significantly with Agreeableness ($r = 0.082$), Conscientiousness ($r = 0.206$), and Openness ($r = 0.081$), although two of the correlations are relatively weak. As expected, Conscientiousness was the most noteworthy predictor of academic performance. It is worth noting that extraversion is sensitive to moderating variables. There was a good degree of consistency among the measurement tools except when measuring Extraversion. Extraversion shows a negative association with academic performance in individualistic contexts ($r = -0.036$). Extraversion is a significant and positive factor for education majors ($r = 0.046$). Proportion of females weakens the influence of Extraversion. In addition, Neuroticism exhibites a significant positive association with performance among freshmen and sophomores ($r = 0.042$). These moderators deepen the understanding of critical role of Big Five theory in explaining academic performance.

1. Introduction

Academic performance is closely related to students career prospects and life well-being (Spengler et al., 2018). Numerous studies investigated the factors that influence academic performance. It was well documented that cognitive ability was one of the robust predictors of academic performance (Leeson et al., 2008; McBride & Greeson, 2023; Preckel et al., 2011). However, there is a notable growing interest in the role of non-cognitive factors in explaining individual differences in academic performance (MacCann et al., 2020; Richardson et al., 2012). One of the non-cognitive factors systematically associated with academic performance is personality (Richardson et al., 2012). The purpose of this study is to clarify the relationship between personality traits, as conceptualized by the Big Five theory (John & Srivastava, 1999), and the academic performance of university students.

Even though there are meta-analyses reporting the effect of Big Five personality on academic performance (Mammadov, 2022; Poropat, 2009; Poropat, 2014), limited studies specifically examine the relationship between Big Five personality and academic performance especially among university students. Meanwhile, several meta-analyses tended to typically concentrate on only one personality trait (e.g., Openness; Gatzka & Hell, 2018), or K-12 education level (e.g., primary education; Poropat, 2014). Vedel's (2014) meta-analysis reported the links between Big Five and tertiary academic performance and his study included 20 research presented prior to the end of 2013. Notably, existing meta-analyses on personality traits and academic performance lack cross-cultural comparisons, with limited data from East Asia. Key factors like measurement tools, diverse majors, year of study as moderators were overlooked, which offer deeper insights into how personality impacts academic outcomes across contextual contexts. This study bridges these gaps using comprehensive, updated data from multiple countries.

2. Literature review

2.1. Big Five theory and academic performance

2.1.1. Big Five theory

Personality traits describe relatively enduring patterns of thoughts, feelings, and behaviors that reflect the tendency to respond in certain

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ways under certain circumstances (Roberts, 2009). Personality traits predict students achievement that an individual can achieve in school or at work (Israel et al., 2022; OECD, 2021.

). Typical attempts to configure systems for capturing personality traits are Cattell's System (Cattell, 1956), Eysenck's System (Eysenck, 1970), Guilford's System (Guilford, 1975), Murray's Need System (Borkenau & Ostendorf, 1989; Costa Jr. & McCrae, 1988), Interpersonal Circle (Leary, 1957) and linguistic analyses (Goldberg, 1981). Digman (1990) summarized that the Big Five personality traits could be categorized into five dimensions, i.e., Agreeableness, Conscientiousness, Extraversion, Neuroticism, Openness. The establishment of the Big Five model of personality has given us a relatively clear framework for describing personality traits (John et al., 2008), effectively avoiding the confusion of words such as temperament, personality, and non-cognitive abilities. According to Saklofske et al.'s (2020) encyclopedia, Openness describes individuals who are intellectually curious, imaginative, creative, and enamored by aesthetic experiences. Conscientiousness refers to a person's tendency toward being self-controlled, persistent, industrious, and orderly. Extraversion implies positive emotions, activity, sociability, and the tendency to seek stimulation in the company of others. Neuroticism means vulnerability to unpleasant emotions such as anxiety, anger, and depression. Graziano and Tobin (2013) define Agreeableness as a set of interrelated dispositions and characteristics, manifested in differences in being likable, pleasant, and harmonious in relation with others. These traits have been extensively studied in various learning and cultural contexts.

2.1.2. Personality traits and academic performance

The linkage between personality traits and academic performance is well-documented (Mammadov et al., 2018). Some researchers focused on the effects of a single dimension of the Big Five personality model on academic performance (Gatzka & Hell, 2018; Sparfeldt & Schwabe, 2024; Thiele et al., 2018), while other studies focused on the effects of multiple or all dimensions of personality traits on academic achievement (Novikova & Vorobyeva, 2017; Zhang & Wang, 2023). As suggested by Mammadov et al. (2018), Conscientious, Openness and Agreeableness were significantly correlated to academic performance, whereas the role of Extraversion and Neuroticism was unstable. Meanwhile, the Big Five personality traits predict academic performance to varying degrees. Conscientiousness was found to be a robust predictor of academic performance across these particular five personality measures in meta-analysis (Mammadov, 2022; Vedel, 2014).

As so far, some meta-analyses provide ample evidence to further reveal the relationship between personality traits and academic performance across different education level (Hessen & Kuncel, 2022; Mammadov, 2022; Poropat, 2009; Trapmann et al., 2007; Vedel, 2014). Including different educational stages may introduce additional heterogeneity and affect the consistency of results. Focusing on the university students could reduce this heterogeneity and improve the accuracy and comparability of the findings. Meanwhile, psychological development and social environments at the university level differ significantly from those at the primary or secondary level, so concentrating on university students can provide more precise insights and interventions.

2.2. Moderators

The second purpose of the present meta-analysis is to examine the influence of moderators on the calculated mean effect sizes. Candidate moderators mainly include personality measurement types, cultural context, academic majors, year of study and gender composition.

2.2.1. Personality measurement types

Scholars developed different scales to measure the Big Five personality. For this meta-analysis, four widely used measures among university students were examined: the Big Five Inventory (BFI; John & Srivastava, 1999); the International Personality Item Pool (IPIP, Goldberg, 1999) or Mini-IPIP (Donnellan et al., 2006); the NEO Five-Factor Inventory (NEO-FFI; Costa & McCrae, 1992) and NEO Personality Inventory-Revised (NEO-PI-R; Costa & McCrae, 1992).

BFI is a 44-item self-report inventory that comprised of adjectives that evaluate the Five Factor Model domains. Mammadov (2022) suggested that the BFI exhibited wide associated heterogeneity across studies. IPIP contains thousands of items, and researchers can create personalized measurement tools by selecting the right combination of items for their needs (Goldberg et al., 2006). The Mini-IPIP scale is a simplified version of the scale based on the IPIP. The Mini-IPIP scale consists of only 20 items, with 4 items for each of the Big Five personality dimensions (Donnellan et al., 2006). NEO-PI-R has a total of 240 items. It is able to provide a more refined picture of the individual, which is particularly suitable for situations where a detailed description of the individual's personality is required, such as academic research, clinical assessment (Piedmont, 1998). The NEO-FFI contains 60 items, 12 items for each of the Big Five personality dimensions, and is a simplified version of the NEO-PI (McCrae & Costa Jr, 2004). The BFI and Mini-IPIP are shorter and easier to administer, making them ideal for large-scale surveys or time-limited contexts. In contrast, the NEO-PI-R provides a more comprehensive and detailed assessment of personality, suitable for in-depth research and clinical use. The NEO-FFI offers a middle ground, balancing detail and efficiency. The IPIP, with its flexibility and open-source nature, allows for greater customization, which can be a significant advantage for researchers who need to adapt their measures to specific populations or study designs. This study attempts to explicate the impact of differences in measurement tools on the relationship between personality traits and academic performance.

2.2.2. Cultural context

It has been pointed out that the sample of studies examining the personality-academic performance relationships is internationally diverse (Schmitt et al., 2007). For example, Trapmann et al.' (2007) meta-analysis identified the correlation Extraversion and the criterion grades separated in four regions (Australia, Europe, North America, East Asia). Mammadov's (2022) meta-analysis grouped evidence from different countries into six regions, namely Asia, Australia, Eastern Europe and Russia, Middle East, North America and Western Europe. Some cultures may place a higher value on hard work and responsibility, and Conscientiousness may be particularly important in academic performance in these regions (Van de Vijver & Leung, 2021). Mammadov's (2022) meta-analysis confirmed that Asian samples consistently showed stronger average correlations between personality traits and academic performance compared to studies from other regions. It suggests that regional cultural differences should be considered as moderating variables.

According to Hofstede (2001) cultural theory, collectivist cultures, with their emphasis on teamwork, social responsibility, and group goals (Calabuig et al., 2018; Triandis, 2018), may place greater value on personality traits such as cooperativeness and agreeableness, which may lead to enhanced positive associations between Agreeableness and academic achievement (Tan et al., 2021). On the other hand, individualistic cultures emphasize personal independence and achievement, often highlighting Openness and responsibility as key traits for reaching personal goals, which aligns with their role in personal success (Heine et al., 2001; Hofstede, 2001; Kormi-Nouri et al., 2015; Oyserman et al., 2002). Nevertheless, there is no meta-analysis incorporating moderating variables from a cultural perspective of collectivism and individualism. Cultural dimensions provide a more precise explanation of behavioural motivations and social norms, whereas regional divisions may have confounding effects due to economic, educational system, political and other factors. The present study regards cultural context as an important variable, facilitating the results to be more comparable globally.

2.2.3. Academic major

Recent research indicates that the influence of personality traits on student performance is significantly shaped by the academic subject (Brandt et al., 2020; Hübner et al., 2022; Meyer et al., 2019). In meta analysis, Vedel (2014) indicated that academic major moderated the relationship between Conscientiousness and GPA. Extraversion may positively affect academic performance in majors that require frequent collaboration and communication, while the effect of this trait may not be significant in majors that place more emphasis on individual research and independent work (De Raad & Schouwenburg, 1996). In some highly structured disciplines such as engineering or accounting, the impact of Openness may be more limited (Ackerman & Heggestad, 1997). Noftle and Robins (2007) suggested that Openness was typically associated with a stronger verbal orientation and a weaker numerical focus. Meyer et al. (2019) identified stronger correlations between Openness and achievement in English subject compared to math in secondary education. Learning in university occurs across various academic majors, academic-specific achievement not only influences students' academic expectations and interests but also determines their future career path (Guo et al., 2017), which makes the inclusion of major moderating variable necessary. In conclusion, scarce meta-analysis so far has specifically addressed associations between Big Five personality traits and university students academic performance with moderator of academic major. Thus, this study hopes to find out whether academic major moderates its relationship.

2.2.4. Year of study

Trapmann et al.' s (2007) meta-analysis reveals that the relationship between Big Five personality traits and academic performance is moderated by the level of study, with Conscientiousness being a stronger predictor for undergraduates than for postgraduates, while other traits like Openness may have a more significant impact at higher year of study. Established meta-analyses generally revealed differences between broader grade levels, i.e. between primary, secondary and tertiary (Mammadov, 2022; Meyer et al., 2019; Poropat, 2009). However, our study hopes to reveal the precise heterogeneity between groups of undergraduate students, which could provide a more nuanced understanding of how personality traits interact with academic performance over time.

2.2.5. Gender composition

Gender differences in personality may help explain gender differences in academic performance (Hicks et al., 2008). Spinath et al.'s (2014) study identified that gender differences in students' individual characteristics varied from non-existent (e.g., general intelligence) to strong (e.g., self-discipline). They elaborated that girls outperformed boys in school, in part because of their greater self-discipline and agreeableness. In addition, traits such as Neuroticism and Extraversion may show different effects depending on gender differences. For example, some studies demonstrated that females were more susceptible to academic stress when they were less emotionally stable, which could affect academic performance (Poropat, 2009). Despite the limited studies, gender appears as a potential moderator to be tested.

2.3. Prior research and present study

2.3.1. Prior meta-analyses

There has been a long history of interest in the relationship between the Big Five personality and university students academic performance. Trapmann et al.'s (2007) article was one of the first meta-analytic studies comprehensively discussing the correlation between the Big Five personality in higher education students' academic performance. Subsequently, Poropat (2009), Richardson et al. (2012), McAbee and Oswald (2013), Vedel (2014), Mammadov (2022) conducted ongoing follow-up studies and reported the combined effect sizes of the Big Five personality affecting academic performance, as shown in the Table 1.

Table 1
Prior meta-analyses of personality traits and academic performance in university students.

Authors	Research focus	Personality traits	Effect size
Trapmann et al., 2007	Tertiary education	Agreeableness	0.059
		Conscientiousness	0.269
		Neuroticism	−0.003
		Extraversion	−0.059
		Openness	0.133
Poropat, 2009	Most come from tertiary education	Agreeableness	0.07
		Conscientiousness	0.22
		Neuroticism	0.02
		Extraversion	−0.01
		Openness	0.12
Richardson et al., 2012	Tertiary education	Agreeableness	0.06
		Conscientiousness	0.23
		Neuroticism	0.01
		Extraversion	−0.03
		Openness	0.09
McAbee & Oswald, 2013	Tertiary education	Agreeableness	0.08
		Conscientiousness	0.23
		Neuroticism	−0.03
		Extraversion	−0.03
		Openness	0.08
Vedel, 2014	Tertiary education	Agreeableness	0.08
		Conscientiousness	0.26
		Neuroticism	0
		Extraversion	−0.01
		Openness	0.07
Mammadov, 2022	69 % of students in tertiary education	Agreeableness	0.09
		Conscientiousness	0.27
		Neuroticism	−0.02
		Extraversion	0.01
		Openness	0.16

Considering more stringent inclusion criteria and the different analyses of moderating variables, this study can be regarded as an innovative meta-analysis. In terms of the inclusion criteria, this paper includes sample size among the criteria for literature screening, which means that some studies with low statistical validity will be excluded, which to some extent can enhance the credibility of the final results (Cohen, 1988; Erdfelder et al., 1996). From the analysis of moderating variables, only Mammadov (2022) has developed a more detailed discussion on education level, personality measurement types, geographical regions of studies, gender, age, publication status heterogeneity. The present study promotes it by adding year of study, academic majors and cultural context on the relationship between personality and academic performance.

2.3.2. The present study

After literature review, we identified the following research gaps. First, almost no meta-analysis compared national perspectives to examine the variability in the impact of students' personality traits on academic performance in different national cultures. Second, existing meta-analyses were predominantly based on evidence from the United States and the United Kingdom (e.g., Mammadov, 2022), with relatively limited sample from East Asian contexts. Third, diverse majors were not used as moderating variables. Fourth, no existing study employed university students' year of study level as a moderator. Accordingly, adopting meta-analysis method, this study attempts to explore the combined effect sizes of the Big Five personality traits on university students' academic performance, and to analyse effect sizes across studies to reveal the sources of heterogeneity. In specific, the present study aims to address three main questions:

QR1: What is the overall strength of the relation between Big five personality traits and academic performance?

QR2: What is the relationship between each dimension of Big five personality traits (i.e., Agreeableness, Conscientiousness, Extraversion, Neuroticism and Openness) and academic performance, respectively?

QR3: Does the relation between personality traits and personality traits vary as a function of moderators of measurement types, cultural context, academic major, year of study, gender composition, respectively?

3. Method

3.1. Literature search

Fig. 1 presents an overview of the literature search process in accordance with the PRISMA guidelines (Tricco et al., 2018). A search of the following databases was conducted in order to identify relevant articles for this review: ERIC, Web of Science databases, PsycINFO, and Scopus. The database search was performed using a combination of key terms, including (a) at least one personality-related keyword indicating that the Big Five was used, which included any personality trait (e.g., Openness, Conscientiousness) or a common measure name (e.g., NEO, Big Five, BFI, etc.); (b) the word personality or emotion intelligence; (c) academic performance or a related term (academic performance, academic achievement, etc.); and (d) university students or synonyms (undergraduate students, freshmen, etc.). The search terms were applied to article titles, abstracts, and search headings. Meanwhile, 54 references meeting inclusion criteria from key meta-analyses on personality and performance-related outcomes were used (Poropat, 2009; Vedel, 2014; Mammadov, 2022).

Overall, the search yielded 11,249 articles. Obvious duplicates (n = 881) were removed after merging these sources. An additional 9,952 articles were excluded at the title and abstract levels due to their irrelevancy to the present study. In the end, it resulted in the retention of 30 articles for further examination for eligibility and inclusion. Searches were conducted in August 2024 and included the years 1995 to 2024. Totally, 84 studies were included in the current meta-analysis. A summary of the studies included in the meta-analysis is presented in Appendix A.

3.2. Inclusion criteria

The criteria for inclusion in the meta-analysis, in conjunction with the requirements of the meta-analysis methodology and the research topic, were:

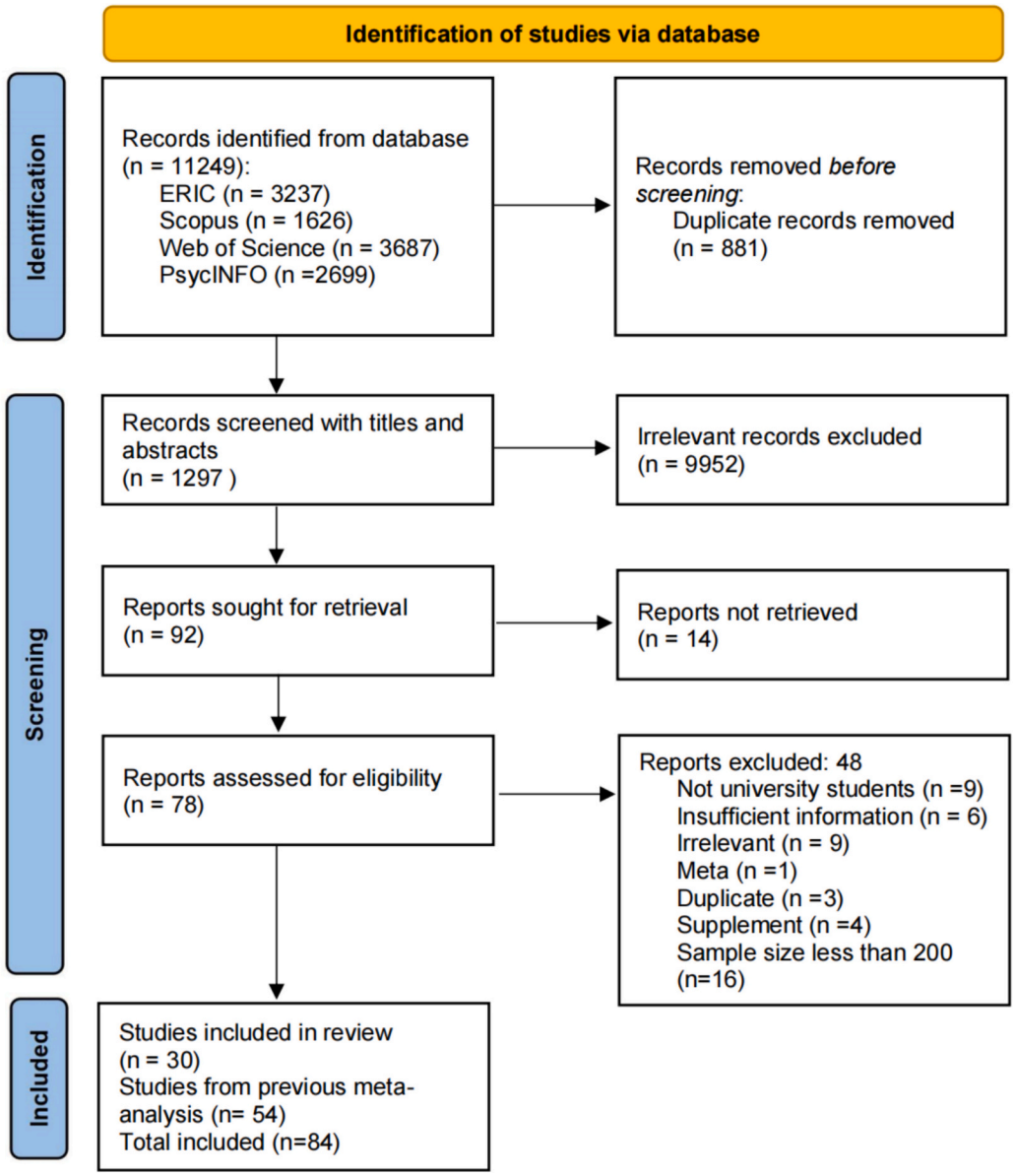

Fig. 1. Flow diagram of literature search process.

(1) Studies were included if they measured personality traits and academic performance using scales that yielded quantitative values, measured Big Five personality traits, measured academic performance through grades, exam performance, GPA, or standardized performance tests;
(2) The study reported at least one correlation (r) between personality traits and academic performance;
(3) Sample size >200.
(4) Be published in a professional journal between 1995 and 2024;
(5) Reported the correlations between each dimension of personality traits and academic performance as effect sizes;
(6) Participants are university students, excluding special education, adult education, community colleges and vocational-technical schools. The focus should be on traditional higher education settings.
(5) Published in English.

According to Cohen (1988), sample size is closely related to statistical power, the larger the sample size, the higher the statistical power. When the sample size is 200, the statistical validity will be more assured (Cohen, 1988; Erdfelder et al., 1996). Therefore, sample size of 200 was used as the inclusion criteria for this study, and studies with a sample size of <200 were excluded. As for publication dates, some meta-analyses emphasized the cut-off date of the literature search in the inclusion criteria (Durlak et al., 2011; Murano et al., 2020), but the start of the search date was ambiguous. The earliest study retrieved in this study that met the inclusion criteria was a study from 1995 (Wolfe & Johnson, 1995).

3.3. Data extraction and coding

The features extracted from each study included authors, publication year, sample size, gender (proportion female), grade, country of sample, personality measurement, academic majors of sample, and correlation coefficients. Percentage of female participants in each of the original studies was used to code gender. Personality measurement was coded as a categorical variable with 6 levels: BFI, IPIP, Mini-IPIP, NEO-FFI, NEO-PI-R, TIPI, Mini-Markers.Cultural context was categorized as collectivism or individualism according to countries. When an article included multiple relevant correlations from different subgroups (e.g., Novikova & Vorobyeva, 2017), we selected independent samples with larger sample sizes. To further refine the coding scheme, a subsample of the data (k = 20) was coded independently by all authors. Interrater agreement levels for the coding categories ranged from 89 % for the methodology section to 100 % for the names of the coding form. The coders subsequently met to compare their results and discuss any discrepancies until reached an agreement upon a final score.

3.4. Meta-analytical procedure

3.4.1. Computation of effect size

We processed the data using Comprehensive Meta-Analysis (Version 3) software. The correlation coefficients were used as effect sizes to integrate the relationship between personality traits and academic performance. The correlation coefficient r values were then converted to Fisher-Z values for meta-analysis (Hedges & Olkin, 1985). Correlation coefficients overestimate the overall effect size because they take values between - 1 and 1. As the correlation coefficient approaches - 1 or 1, the distribution becomes more skewed. Overestimation also arises from the nonlinear relationship between correlation coefficients and Fisher-Z values, particularly as correlations approach extreme values. To address this problem, the correlation coefficients were converted to Fisher-Z and weighted according to the inverse of the variance in order to give greater weight to larger samples than smaller ones (Lipsey & Wilson, 2001). Mean effect sizes were then obtained by transforming from *Z*-scores to correlation coefficients, r, with confidence intervals to indicate the range into which the overall mean might fall in the observed data (Hedges & Olkin, 1985). The transformation from sample correlation r to Fisher's z is given by $z = 0.5 \times ln\left(\frac{1+r}{1-r}\right)$, then Fisher's z could be converted back to correlation using $r = \frac{e^{2z}-1}{e^{2z}+1}$ (Borenstein et al., 2009; Lipsey & Wilson, 2001). Mean effect sizes and confidence intervals were converted back to correlation coefficients for presentation (Lipsey & Wilson, 2001). Effect sizes were weighted by the inverse sampling variance.

3.4.2. Heterogeneity test

In calculating overall effect sizes, there are usually two types of statistical models, fixed-effects models and random-effects models. The former assumes that the studies included in the analysis are homogeneous, that differences in observed effect sizes are attributable to sampling error, and that only the aggregate covered by the study is included and cannot be generalized to other aggregates. The latter assumes that the studies included in the analysis are not homogeneous in terms of study populations and research instruments, and therefore we should not assume that they share a common effect (Schmidt et al., 2009). The literature review of this study found that the relationship between Big Five personality traits and academic achievement may be influenced by factors such as gender, measurement instruments, and geographic culture of the subject group. When different study characteristics can affect the results of meta-analysis, it is more reasonable to choose a random effects model (Borenstein et al., 2009).

The rationality of random effects model selection is further verified by heterogeneity test. The heterogeneity test methods mainly include Q-test and I^2 test. The Q-test is a test based on the total variation, assuming that the effect sizes follow a chi-square distribution, which indicates significant heterogeneity if $p < 0.05$ (Cochran, 1954). The I^2 test mainly reflects the proportion of the true variance of the effect size in the total variance (Higgins & Thompson, 2002). I^2 values of 25 %, 50 % and 75 % can be considered as boundaries of low, medium and high heterogeneity (Higgins et al., 2003). Meanwhile, tau-squared (τ^2), defined as the variance of the true effect sizes was reported. The larger $\tau,^2$ the greater the difference in true effects among different studies, indicating stronger heterogeneity (Borenstein et al., 2009). If heterogeneity is found in a meta-analysis, meaning there is considerable variation in effect sizes across studies, researchers typically introduce moderators to examine which factors may explain this difference (Borenstein et al., 2009).In this study, personality measurement types, cultural context served as moderators, and academic major, year of study, and gender composition were regarded as moderators. In addition, meta-regression was used for non-categorical moderators (gender composition) to test if these variables were significant covariates within the meta-regression model.

3.4.3. Publication bias

Publication bias refers to the issue of selective reporting, where studies with significant findings are more likely to be accepted and published (Rosenthal, 1979). We assessed publication bias using two methods. First, we applied Egger's regression test to examine the asymmetry of funnel plots (Egger et al., 1997; Pustejovsky & Rodgers, 2019). Second, we utilized contour-enhanced funnel plots to detect publication bias (Peters et al., 2008). Studies with non-significant results are expected to fall outside the significance contours, and asymmetry within these contours suggests publication bias, while asymmetry outside these contours may be attributed to factors like heterogeneity. If asymmetry is detected, we applied the trim-and-fill method to investigate the potential missing studies and the results were used to adjust the reported effect sizes (Duval & Tweedie, 2000). However, since no publication bias or funnel plot asymmetry was detected, the results of the trim-and-fill method did not adjust the effect sizes nor alter our final conclusions.

4. Results

4.1. Study characteristics

A total of 84 articles were included in this study, with an independent effect size of 370 and a sample size distribution between 201 and 10,497, totalling 46,729 individuals. Sample from 26 different countries or regions. 68 studies reported the five dimensions of the Big Five personality (80.95 %), while the remaining 16 reported only some of the dimensions. Most of the literature reported in some detail sample characteristics, such as gender (88.10 %), year of study (99.81 %), majors (99.81 %), and country or region of affiliation (100 %). All the articles reported instrument for measuring Big Five personality. It laid the groundwork for subsequent analyses of moderating variables. The basic characteristics of the literature, sample characteristics, and some statistical information of the included literature were shown in the Appendix 1.

4.2. Meta-analytic correlations

Table 2 gave the results of meta-analysis of Big Five personality traits and academic performance. The overall correlation between personality traits and academic performance was $r = 0.092$, 95 % CI [0.063, 0.122]. The effect sizes of Agreeableness ($r = 0.082$, 95 % CI [0.050, 0.113]), Conscientiousness ($r = 0.206$, 95 % CI [0.170, 0.241]), and Openness ($r = 0.081$, 95 % CI [0.055, 0.108]) on academic achievement were all significant. Thus, Conscientiousness showed a higher correlation with academic performance followed by Openness. However, the 95 % confidence intervals of Extraversion and Neuroticism included zero, suggesting that their correlations with performance were not significant.

Tests of the homogeneity of the effect sizes using Q statistic (Hedges & Olkin, 1985) were conducted. For each personality trait, the Q value was statistically significant, ranging from 471.431 to 1141.673, suggesting differences in effect sizes across samples. The large I^2 heterogeneity index ranged from 84.235 % to 92.993 %, indicating that an overwhelming amount of observed variation between samples was due to systematic between-samples variability. Therefore, subgroup analyses and meta-regression were performed to analyse potential moderators and identify sources of heterogeneity.

4.3. Moderator analysis

4.3.1. Moderating effects of personality measurement types

A series of subgroup analyses were performed to examine the moderating effect of personality measurement types (BFI, IPIP, NEO-FFI, and NEO-PI). As presented in Table 3, there was a good degree of consistency among the measures. Agreeableness showed a positive correlation with academic performance. Further, the most consistent meta-analytic correlations across scales were observed in Conscientiousness. The effect sizes ranged from 0.190 to 0.219. However, when measuring Conscientiousness by BFI, it exhibited the highest heterogeneity (Q = 264.854, I^2 = 93.581 %), suggesting substantial differences across studies using BFI. In contrast, Extraversion showed weaker and more inconsistent relationships. Neuroticism generally had a negative but non-significant relationship with academic performance across tools. Openness and academic performance exhibited non-significant correlation when adopting BFI.

Table 3
Subgroup analysis of personality–academic performance relationships by personality measurement types.

	k	r	95 % CI	p	Q	I^2	τ^2
Agreeableness							
BFI	15	0.062	[0.006, 0.118]	0.031	98.520	85.790	0.010
IPIP	15	0.069	[0.008, 0.131]	0.028	68.580	79.586	0.012
NEO-FFI	19	0.103	[0.066, 0.140]	0.000	44.072	59.158	0.004
NEO-PI	8	0.091	[0.009, 0.173]	0.030	40.494	82.713	0.012
Conscientiousness							
BFI	18	0.219	[0.196, 0.224]	0.000	264.854	93.581	0.021
IPIP	18	0.190	[0.147, 0.232]	0.000	51.223	66.812	0.006
NEO-FFI	20	0.199	[0.127, 0.268]	0.000	191.335	90.070	0.025
NEO-PI	8	0.217	[0.145, 0.285]	0.000	31.728	77.938	0.009
Extraversion							
BFI	15	−0.049	[−0.079, −0.019]	0.002	25.084	44.188	0.001
IPIP	15	−0.015	[−0.057, 0.027]	0.494	31.239	55.184	0.004
NEO-FFI	18	−0.041	[−0.087, 0.004]	0.004	61.114	72.183	0.007
NEO-PI	8	0.001	[−0.092, 0.094]	0.989	51.158	86.317	0.015
Neuroticism							
BFI	15	−0.018	[−0.062, 0.027]	0.444	61.302	77.162	0.005
IPIP	16	−0.071	[−0.207, 0.067]	0.313	390.966	96.163	0.077
NEO-FFI	19	−0.033	[−0.102, 0.036]	0.343	156.270	88.481	0.020
NEO-PI	8	−0.055	[−0.141, 0.031]	0.207	43.739	83.996	0.013
Openness							
BFI	14	0.059	[−0.005, 0.123]	0.069	112.716	88.467	0.012
IPIP	15	0.061	[0.017, 0.105]	0.006	34.045	58.878	0.004
NEO-FFI	20	0.111	[0.082, 0.140]	0.000	29.359	35.284	0.001
NEO-PI	8	0.062	[0.016, 0.108]	0.008	12.562	44.276	0.002

4.3.2. Moderating effect of cultural context

As shown in Table 4, combining Hofstede's (2001) individualistic and collectivistic cultural framework, the present study compared results across different cultural contexts. In terms of Agreeableness, in individualistic contexts, Agreeableness showed a slight significant positive correlation with academic performance ($r = 0.059$, 95 % CI [0.034, 0.084]). In collectivistic contexts, the effect size was higher ($r = 0.126$,

Table 2
Main meta-analysis correlations.

Domain	k	r	95%CI	Q	I^2	τ^2
Agreeableness	72	0.082	[0.050, 0.113]	673.030	89.451	0.016
Conscientiousness	81	0.206	[0.170, 0.241]	1141.673	92.993	0.026
Extraversion	72	−0.009	[−0.036, 0.017]	471.431	84.939	0.011
Neuroticism	73	−0.029	[−0.065, 0.008]	900.231	92.002	0.022
Openness	72	0.081	[0.055, 0.108]	450.357	84.235	0.010
Overall	84	0.092	[0.063, 0.122]	803.583	89.671	0.017

Notes: k, number of independent studies; r, estimated correlation coefficient; 95 % CI, 95 % confidence intervals for estimated correlation coefficient; Q, Cochran's measure of heterogeneity; I^2, Higgins' measure of heterogeneity; τ^2 between-study sampling variance.

Table 4
Subgroup analysis of personality–academic performance relationships by cultural context of studies.

	k	r	95 % CI	p	Q	I^2	τ^2
Agreeableness							
Individualistic	42	0.059	[0.034, 0.084]	0.000	114.659	64.242	0.004
Collectivistic	27	0.126	[0.049, 0.202]	0.001	451.614	94.243	0.039
Conscientiousness							
Individualistic	50	0.209	[0.174, 0.244]	0.000	333.751	85.318	0.014
Collectivistic	27	0.205	[0.106, 0.299]	0.000	789.355	96.706	0.069
Extraversion							
Individualistic	42	−0.036	[−0.065, −0.005]	0.022	157.992	74.049	0.007
Collectivistic	27	0.024	[−0.031, 0.079]	0.384	228.194	88.606	0.018
Neuroticism							
Individualistic	44	0.006	[−0.030, 0.041]	0.755	245.858	82.510	0.011
Collectivistic	26	−0.090	[−0.183, 0.005]	0.063	624.439	95.996	0.058
Openness							
Individualistic	42	0.067	[0.038, 0.095]	0.000	147.478	72.199	0.006
Collectivistic	27	0.101	[0.037, 0.164]	0.002	280.561	90.733	0.026

95 % CI [0.049, 0.202]), suggesting that cultural factors might amplify this trait' s influence on academic performance. Conscientiousness was a robust predictor of academic performance in both individualistic (r = 0.209, 95 % CI [0.174, 0.244]) and collectivistic contexts (r = 0.205, 95 % CI [0.106, 0.299]). As for Extraversion, it exhibited a negative relation with academic performance in individualistic contexts (r = −0.036, 95 % CI [−0.065, −0.005]). In collectivistic contexts, no significant relationship was found. With respect to Neuroticism, it had no statistically significant correlation with academic performance in collectivistic context. Openness was positively associated with academic performance in both individualistic (r = 0.067, 95 % CI [0.038, 0.095]) and collectivistic contexts (r = 0.101, 95 % CI [0.037, 0.164]), although the effect was larger in collectivistic settings.

4.3.3. *Moderating effects of academic majors*

In Table 5, the study examined whether the relationship between the Big Five personality traits and academic performance varied across different academic majors. As for Agreeableness, for psychology students, the effect size was smaller but statistically significant (r = 0.055, 95 % CI [0.003, 0.096]). However, for other majors, there was no significant effect was revealed. On contrast, Conscientiousness showed a consistent positive relationship with academic performance across all majors and correlations ranged from 0.178 to 0.228. In terms of Extraversion, the effect size of it was significant and positive for education majors (r = 0.046, 95 % CI [0.004, 0.088]), whereas there was no significant correlation between Extraversion and academic performance across other majors. Regarding to Neuroticism, there was generally a negative relationship with academic performance, although not all findings were statistically significant. Openness consistently demonstrated a significant positive correlation to academic performance across most majors except medical. Conscientiousness consistently showed a higher and positive relationship across all majors, while Agreeableness and Extraversion indicated more variability in their effects depending on the major.

4.3.4. *Moderating effects of year of study*

In Table 6, the meta-analysis investigated the relationship between the Big Five personality traits and academic performance, with year of study serving as a moderating variable. The data was divided into two subgroups: freshmen and sophomores, and juniors and seniors. First, Agreeableness showed a small, significant and similar positive relationship with academic performance between two groups. Second, Conscientiousness showed a significant positive associations with academic performance among freshmen and sophomores (r = 0.135, 95 %

Table 5
Subgroup analysis of personality–academic performance relationships by academic majors.

	k	r	95 % CI	p	Q	I^2	τ^2
Agreeableness							
Education	7	0.055	[−0.023, 0.132]	0.170	30.280	80.185	0.009
Management & Business	6	0.116	[−0.039, 0.265]	0.141	67.625	92.606	0.035
Psychology	13	0.055	[0.003, 0.096]	0.008	29.047	58.688	0.003
Medical	6	0.091	[−0.085, 0.262]	0.311	85.406	94.146	0.046
Conscientiousness							
Education	6	0.186	[0.073, 0.294]	0.001	37.152	86.542	0.018
Management & Business	8	0.178	[0.041, 0.308]	0.011	92.151	92.404	0.037
Psychology	17	0.228	[0.167, 0.287]	0.000	125.879	87.289	0.015
Medical	6	0.220	[0.028, 0.397]	0.025	105.062	95.241	0.057
Extraversion							
Education	7	0.046	[0.004, 0.088]	0.031	8.858	32.266	0.001
Management & Business	6	0.026	[−0.095, 0.147]	0.676	41.481	87.946	0.020
Psychology	13	−0.043	[−0.091, 0.006]	0.086	42.242	71.592	0.005
Medical	6	−0.040	[−0.111, 0.032]	0.276	13.707	63.523	0.005
Neuroticism							
Education	7	−0.024	[−0.090, 0.042]	0.476	21.718	72.373	0.006
Management & Business	8	−0.235	[−0.486, 0.053]	0.108	317.475	98.110	0.152
Psychology	14	−0.039	[−0.106, 0.028]	0.253	93.196	86.051	0.014
Medical	6	−0.024	[−0.139, 0.092]	0.685	36.570	86.328	0.018
Openness							
Education	6	0.125	[0.039, 0.210]	0.005	21.221	76.438	0.009
Management & Business	6	0.080	[0.019, 0.139]	0.010	10.111	50.549	0.003
Psychology	14	0.071	[0.007, 0.134]	0.031	90.962	85.708	0.012
Medical	6	0.154	[−0.009, 0.309]	0.064	73.415	93.189	0.039

Table 6
Subgroup analysis of personality-academic performance relationships by year of study.

	k	r	95 % CI	p	Q	I^2	τ^2
Agreeableness							
Freshmen & sophomores	20	0.055	[0.017, 0.093]	0.005	59.057	67.828	0.005
Juniors & seniors	13	0.069	[0.021, 0.116]	0.005	29.844	59.790	0.005
Conscientiousness							
Freshmen & sophomores	26	0.135	[0.068, 0.201]	0.000	326.014	92.332	0.028
Juniors & seniors	13	0.210	[0.142, 0.276]	0.000	65.007	81.541	0.013
Extraversion							
Freshmen & sophomores	20	−0.019	[−0.066, 0.029]	0.439	92.641	79.491	0.009
Juniors & seniors	13	−0.019	[−0.092, 0.054]	0.616	70.643	83.013	0.015
Neuroticism							
Freshmen & sophomores	22	0.042	[0.000, 0.083]	0.048	86.679	75.773	0.007
Juniors & seniors	13	−0.054	[−0.119, 0.012]	0.108	57.177	79.012	0.011
Openness							
Freshmen & sophomores	21	0.062	[0.028, 0.095]	0.000	51.556	61.207	0.004
Juniors & seniors	13	0.069	[0.017, 0.120]	0.009	35.118	65.830	0.006

CI [0.068, 0.201]). The relationship was higher for juniors and seniors ($r = 0.210$, 95 % CI [0.142, 0.276]). This implied that Conscientiousness, a trait tied to self-discipline and responsibility, became more predictive of academic performance as students moved into upper years. Third, Extraversion did not significantly predict academic performance for both groups. Fourth, Neuroticism had a small but significant positive relationship with academic performance among freshmen and sophomores ($r = 0.042$, 95 % CI [0.000, 0.083]). Fifth, Openness showed a positive relationship with academic performance for both freshmen and sophomores ($r = 0.062$, 95 % CI [0.028, 0.095]) and juniors and seniors ($r = 0.069$, 95 % CI [0.017, 0.120]), suggesting Openness was a stable predictor of academic success across different year of study.

4.3.5. Meta-regression for gender composition

Table 7 presented the results of moderator analyses using meta-regression for gender composition. Percentage of females served as a continuous variable when testing moderating effects of gender distribution. Studies included in the meta-analysis tended to have more females than males. The mean percentage of females was 62.6 %.

The intercepts for each personality trait were significant in most cases except for Openness, suggesting that these traits were related to academic performance, even when gender distribution was not considered. However, the proportion of females demonstrated varying degrees of moderating effects across the traits. For Agreeableness ($\beta = 0.229$, CI [0.077, 0.381], $p < 0.01$) and Conscientiousness ($\beta = 0.329$, CI [0.154, 0.504], $p < 0.001$), both showed a positive baseline relationship with academic performance, but the proportion of females had a non-significant moderating effect. Extraversion had a positive baseline effect ($\beta = 0.143$, CI [0.065, 0.016], $p < 0.05$), with a significant negative moderation by female proportion ($\beta = -0.235$, CI [−0.434, −0.035], $p < 0.05$). Neuroticism presented a negative baseline effect ($\beta = -0.197$, CI [−0.374, −0.019], $p < 0.05$), without moderation by female proportion. The associations between Openness and academic performance did not appear to be affected by gender distribution.

4.4. Publication bias

A publication bias analysis was conducted to examine whether studies with statistically significant correlations were more likely to be published than non-significant results. The analysis of weighted random effects model in which the publication status was included as a dichotomous moderator suggested a minimal concern. As shown in AppendixA, the funnel plot appeared to be symmetrical. This seemed to imply that there was no obvious problem of publication bias in this study.

Another more accurate way used to assess publication bias was the Egger's regression intercept test. With Egger's test, we found no significant asymmetry for overall studies, $r = 0.092$, 95 % CI = [−0.671, 2.305]. As for different personality traits, Agreeableness ($r = 0.082$, 95 % CI = [−0.422, 2.280]), Conscientiousness ($r = 0.206$, 95 % CI = [−2.155, 1.522]), Extraversion ($r = -0.009$, 95 % CI = [−1.296, 1.311]), Neuroticism ($r = -0.029$, 95 % CI = [−3.429, 0.019]), and Openness ($r = 0.081$, 95 % CI = [−0.846, 1.687]). There was no need to apply the Trim-and-Fill method. In summary, it suggested that the Egger's test did not find significant publication bias in the overall study and in the meta-analyses of each personality trait. The results of the study were reliable.

5. Discussion

The present meta-analysis provides a comprehensive assessment of the links between university students self-reported personality traits and academic performance from 84 studies with a total of 370 correlations and an aggregated sample size of 46,729. It provides more reliable and valid estimates compared with previous similar meta-analyses, as it includes substantially more studies and moderators to explain heterogeneity.

Table 7
Meta-regression testing moderating effects of gender distribution.

Moderator	k	β	SE	95%CI		Z	p
				Lower	Upper		
Agreeableness (intercept)	61	0.229	0.078	0.077	0.381	2.95	0.003
Proportion female		−0.225	0.121	−0.462	0.012	−1.86	0.063
Conscientiousness (intercept)	70	0.329	0.089	0.154	0.504	3.68	0.000
Proportion female		−0.167	0.140	−0.441	0.107	−1.19	0.233
Extraversion (intercept)	61	0.143	0.065	0.016	0.270	2.30	0.028
Proportion female		−0.235	0.102	−0.434	−0.035	−2.30	0.021
Neuroticism (intercept)	62	−0.197	0.091	−0.374	−0.019	−2.17	0.030
Proportion female		0.273	0.141	−0.004	0.550	1.93	0.054
Openness (intercept)	61	0.097	0.067	−0.034	0.228	1.45	0.147
Proportion female		−0.014	0.104	−0.219	0.190	−0.14	0.891

5.1. Big Five personality traits and academic performance

Overall, results demonstrated that three of the Big Five personality traits, Agreeableness ($r = 0.082$), Conscientiousness ($r = 0.206$), Openness ($r = 0.081$) showed significant and positive associations with academic performance. Vedel's (2014) and Mammadov's (2022) found that GPA was correlated significantly with Agreeableness, Conscientiousness, and Openness, even though Mammadov studied students at all educational levels.

As expected, Conscientiousness was clearly the prominent personality predictor of academic performance ($r = 0.208$). This finding is consistent with previous studies indicating that Conscientiousness was a robust predictor of tertiary GPA (Trapmann et al., 2007; Vedel, 2014). The estimate was slightly larger than that the corrected mean effect sizes reported in Poropat (2009; $r = 0.19$, $N = 70{,}926$) study, whose sample mostly came from the tertiary level of education. But this estimate was slightly smaller than effect size reported by McAbee and Oswald (2013; $r = 0.26$, $N = 26{,}382$), and Vedel (2014; $r = 0.26$, $N = 17{,}717$). Both of them concentrated on postsecondary students GPA and adopted meta-analysis method, collecting literature from 1993 to 2013. Individuals with high conscientiousness are more likely to plan, set goals, and stick to schedules, which impacts their time management and achieve academic success (Britton & Tesser, 1991). As for Agreeableness, it emerged as a significant and positive predictor of academic performance ($r = 0.084$). This correlation was similar to Mammadov's (2022) finding ($r = 0.09$) and slightly larger than O'Connor and Paunonen's (2007) result ($r = 0.06$). Based on social support theory, individuals with high agreeableness are better at establishing relationships and collaborating with others, which may make it easier to receive social support (Mishra, 2020). Positive teacher-student relationships, peer support, and effective teamwork can contribute to academic performance. Besides, Openness demonstrated significant positive association with academic performance ($r = 0.082$). It was smaller than those reported in Poropat's (2009) meta-analysis. Openness is closely linked to cognitive flexibility and ability to resolve complex problems, which contributes to academic success (Leeson et al., 2008).

Regarding to Neuroticism, the correlation between Neuroticism and academic performance was −0.029 without significance. Vedel's (2014) and Mammadov's (2022) revealed the correlation between Neuroticism and GPA was not statistically significant with $r = -0.02$ as well. Trapmann et al. (2007) and Richardson et al. (2012) also indicated that Neuroticism did not an important predictor of academic grades. The present study is consistent with their findings. However, Poropat (2009) reported a significant association of Neuroticism with performance at tertiary education ($r = -0.246$). In addition, Extraversion was not significantly related to academic performance. Trapmann et al. (2007) also suggested that Extraversion had no significant impact on academic success. Extraversion is associated with social skills, energy, and positive emotional experiences. It may benefits language major but distracts math (Meyer et al., 2019). It is not directly related to academic performance, which is more dependent on an individual's self-discipline, and time management skills, which are more often characterized by Conscientiousness (O'Connor & Paunonen, 2007; Poropat, 2009).

5.2. Moderators analysis

This study unraveled heterogeneity between Big Five personality traits and academic achievement in terms of measurement types, cultural context, academic majors, year of study, and gender composition.

5.2.1. Moderating effects of personality measurement types

In general, consistency is expected among the measures. As McAbee and Oswald (2013) pointed out that although the criterion-related validities for Extraversion, Agreeableness, Neuroticism, and Openness differed by measuring, they were generally low ($r < 0.10$). Although different studies used different scales to measure the Big Five personality traits, correlations between scales on the dimension of Conscientiousness showed consistency, i.e., results were relatively stable across studies. This might because Conscientiousness, as a stable personality trait, is not susceptible to changes in context or measurement methods. In contrast, Extraversion and Openness showed inconsistent relationships across different measurement tools. Extraversion typically includes a number of aspects such as sociality, vitality, and positive affect, and different measurement instruments may emphasize or interpret these aspects with varying emphasis, leading to variability in results when measuring Extraversion (DeYoung et al., 2007; Smillie et al., 2012; Soto & John, 2017). Another notable result was that the BFI exhibited heterogeneity when measuring Openness, and BFI also presented noteworthy heterogeneity in Mammadov's (2022) study. This may be due to differences in the number of items and dimensions across different versions of the BFI, leading to variations in the measurement of Openness, which in turn influences the stability of the correlation.

5.2.2. Moderating effect of cultural context

The results indicated that the cultural context, specifically individualism and collectivism, moderated the relationship between the Big Five personality traits and academic performance. Conscientiousness consistently emerged as a strong predictor across contexts. Meanwhile, Agreeableness, Openness showed stronger and beneficial predictors of academic achievement in collectivist cultures. In contrast, Extraversion' presented a negative effect in individualistic contexts. Markus and Kitayama's (1991) self-construal theory could explain these findings. In collectivist cultures, individuals prioritize group interests, and personality traits such as cooperation with others and adaptability to diverse learning perspectives (e.g., Agreeableness and Openness) contribute to academic performance. However, in individualistic cultures, Extraversion may be closely associated with competitiveness and self-directed behaviors, and excessive Extraversion could lead to self-centeredness and neglect of cooperation, which may hinder academic performance.

Overall, this study is the first meta-analysis to adopt collectivist and individualist cultures as moderating variable, which provides a finer-grained explanation of the cultural context of the study. It transcends geographic and regional limitations and allows for a more comprehensive and globally applicable analysis of the relationship between personality traits and academic performance.

5.2.3. Moderating effects of academic majors

Conscientiousness showed a consistent positive relationship with academic performance across all majors. In striking contrast, the correlation between Neuroticism and academic performance was not significant within all majors. The study also revealed that the positive effect of Agreeableness and Openness on academic performance was significant among psychology students, which was consistent to Vedel's (2014) findings. Notably, the correlation of Extraversion was only significant and positive for education majors, whereas there was no significant correlation among other major students. There were particularly few studies that use major as a moderating variable, and this finding was intriguing. To our knowledge, education may emphasize classroom discussions, experiential teaching, and collaborative teamwork, creating an environment that prioritizes effective communication, interpersonal interaction, and teaching presentation skills. Extroversion, as a personality trait, may facilitate more active engagement in these activities, thereby contributing to improved academic performance. Academic performance in other disciplines typically demands robust theoretical learning and the mastery of specialized skills, with an emphasis on logical reasoning, analytical proficiency, and the accumulation of domain-specific knowledge (Thiele et al., 2018). To sum up, different disciplines have different preferences for personality traits. These findings explore the relationship between personality traits and academic performance from an academic majors perspective at a more refined level than previous meta-analyses.

5.2.4. Moderating effects of year of study

The results suggested that the relationship between personality traits and academic performance was moderated by students' year of study. Agreeableness, Conscientiousness, Openness demonstrated stronger associations with academic success among juniors and seniors. They might participate in more advanced coursework, such as group projects or internships, which attaches great importance on individual quality such as sense of responsibility, collaboration, and critical thinking. Students also need to regulate their behaviors and emotions, and adjust learning strategies to cope with the complex coursework. Interestingly, Neuroticism exerted a significant positive influence on performance among freshmen and sophomores but not juniors and seniors. This pattern suggested that emotional sensitivity and anxiety might serve as motivators in the early stages of university education (Pekrun et al., 2002). As students age, their stress coping mechanisms improves. The need to deal with multiple stresses such as academics and employment could dilute the impact on academic achievement. Extraversion, by contrast, did not significantly influence academic performance in either subgroup. This may reflect the predominantly independent and study-focused nature of higher education, where extroverted behaviors provide limited academic advantage. Openness, on the other hand, consistently predicted positive academic outcomes across grade levels, suggesting that students who were receptive to new ideas and experiences were better equipped to thrive throughout their academic careers (Chamorro-Premuzic & Furnham, 2009).

5.2.5. Meta-regression for gender composition

While Big Five personality traits are related to academic performance, gender distribution, specifically the proportion of females, played a nuanced role in moderating this relationship, particularly weakening the positive effect of Extraversion, while having little to no impact on other traits (e.g., agreeableness, conscientiousness, Neuroticism and Openness). Mammadov (2022) also elaborated that Extraversion was a less effective predictor of academic performance in samples with a higher proportion of female participants. According to Eagly and Wood's (1999) social role theory, societal expectations and norms related to gender roles played a significant role in shaping behavior across different gender groups. In environments with a higher proportion of females, there may be stronger social expectations and norms encouraging students to engage in quieter, more independent study, potentially diminishing the social advantages of Extraversion (Cuddy et al., 2015).

6. Conclusion and implications

6.1. Conclusion

This meta-analysis reaffirmed the significant role of Agreeableness, Conscientiousness and Openness in predicting academic performance among university students, with Conscientiousness emerged as the most robust predictor. Whereas Neuroticism and Extraversion showed little negative correlation to academic outcomes. The study also revealed that measurement tools, cultural contexts, academic majors, year of study and gender composition acted as moderators, providing more comprehensive and interpretive perspectives.

6.2. Implications

From perspective of theoretical contribution, this study not only applied the Big Five theory in higher education research, particularly by analyzing the relationship between academic performance and personality traits across different disciplines, but also acknowledged the situational and cultural dynamics that influenced students outcomes. It extended the practical application of the Big Five theory. In the meantime, the study discussed the intrinsic mechanisms between personality and academic performance from perspectives such as emotional regulation, social support, and learning strategies, which enriched the empirical evidence of the Big Five theory.

This study shed lights on students' personalized education and career planning. Based on our findings, first, personality assessments can offer additional interpretative insights and serve as a valuable supplementary screening method, complementing aptitude and talent evaluations to identify students at risk of academic underachievement or failure. Second, educators are advised to utilize psychological assessment tools to identify students with high levels of neuroticism early in their college journey, such as emotional regulation training and counseling, and tailor interventions or support strategies for students at different academic stages. Third, schools are expected to take into account the cultural background of students when designing the curriculum. For instance, addressing the potential challenges associated with Extraversion by promoting environments that balance social engagement with opportunities for focused, independent work. Last but not least, it should be noted that personality during early-life remains malleable and contributes to students' learning behaviors, motivation, and self-regulation skills, which persist into university years and influence academic performance (Soto & Tackett, 2015). Students are advised to choose educational experiences and environments that matched their personalities (Lüdtke et al., 2011).

7. Limitation and future research

This study offers specialized and comprehensive assessments to date of the relationship between personality traits and academic performance among university students, reaffirming the critical role that personality plays in academic outcomes. However, there are several limitations to consider.

Similar to past meta-analyses, our study focuses on cross-sectional data. Almost all of the studies took the form of self-assessment on personality traits, which may be subject to subjectivity and reactivity. Its relationship with academic performance under other measures or longitudinal study could be further examined. Because of the wide variation in how academic achievement is calculated in each study, although there are basic forms of GPA and exam scores, more precise delineations will be needed in the future to assess the moderating effects of types of academic achievement. In addition, this study did not test for the presence of nonlinear relationships between personality traits and academic performance. If it is hypothesized that various combinations of personality traits may exert nonlinear effects, future research should consider incorporating interaction terms to investigate these intricate dynamics.

CRediT authorship contribution statement

Shuzhen Chen: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Alan Chi Keung Cheung:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Zhaobing Zeng:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.paid.2025.113163.

Data availability

I can provide a link to the raw data at the end of the article

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